**6** A shop sells packs of black pens, packs of red pens and packs of green pens.

There are

2 pens in each pack of black pens
5 pens in each pack of red pens
6 pens in each pack of green pens

On Monday,

$$\begin{matrix}\text{number of packs}\\ \text{of black pens sold}\end{matrix} : \begin{matrix}\text{number of packs}\\ \text{of red pens sold}\end{matrix} : \begin{matrix}\text{number of packs}\\ \text{of green pens sold}\end{matrix} = 7:3:4$$

A total of 212 pens were sold.

Work out the number of green pens sold.

Pens per pack × packs sold:

black 2 × 7 = 14; red 5 × 3 = 15;

green 6 × 4 = 24. Total = 53 parts.

**1 mark**
Convert the pack ratio into pens.

| | Black | Red | Green | Total |
|---|---|---|---|---|
| | 14 | 15 | 24 | 53 |
| ×4 | 56 | 60 | 96 | 212 |

**1 mark**
24 of 53 parts are green.

**1 mark**
Scale 53 parts to 212 pens.

Green pens = 24 × 4 = 96

**1 mark**
96 green pens.

96

**(Total for Question 6 is 4 marks)**